Assessment

Name: Shreya Shruti

UID: 24BCS12997

Sec: 601KRG-B

Sub: Competitive Coding

Q.1. Playlist (CSES)

CSES Problem Set

Playlist

TASK | SUBMIT | RESULTS | STATISTICS | TESTS

Submission details

Task:	Playlist
Sender:	shreya01
Submission time:	2026-04-10 08:30:36 +0300
Language:	C++ (C++20)
Status:	READY
Result:	TIME LIMIT EXCEEDED

cses.fi/problemset/result/16997446/

Language:	C++ (C++20)
Status:	READY
Result:	TIME LIMIT EXCEEDED

Test results

test	verdict	time	
#1	ACCEPTED	0.00 s	»
#2	ACCEPTED	0.00 s	»
#3	ACCEPTED	0.00 s	»
#4	ACCEPTED	0.00 s	»
#5	ACCEPTED	0.00 s	»
#6	ACCEPTED	0.05 s	»
#7	ACCEPTED	0.05 s	»
#8	ACCEPTED	0.06 s	»
#9	ACCEPTED	0.14 s	»
#10	ACCEPTED	0.17 s	»
#11	ACCEPTED	0.14 s	»
#12	ACCEPTED	0.14 s	»
#13	ACCEPTED	0.00 s	»
#14	ACCEPTED	0.07 s	»
#15	ACCEPTED	0.14 s	»
#16	ACCEPTED	0.00 s	»
#17	ACCEPTED	0.14 s	»
#18	ACCEPTED	0.00 s	»
#19	TIME LIMIT EXCEEDED	--	»
#20	ACCEPTED	0.12 s	»
#21	ACCEPTED	0.00 s	»

Code

Code

```
#include <bits/stdc++.h>
using namespace std;

int main()
{
    int n;
    cin >> n;

    vector<int> songs(n);
    for (int i = 0; i < n; i++)
        cin >> songs[i];

    unordered_map<int, int> lastSeen;
    int left = 0, maxLen = 0;

    for (int right = 0; right < n; right++)
    {

        if (lastSeen.count(songs[right]) && lastSeen[songs[right]] >= left)
        {
            left = lastSeen[songs[right]] + 1;
        }

        lastSeen[songs[right]] = right;

        maxLen = max(maxLen, right - left + 1);
    }

    cout << maxLen << endl;
}
```

Q.2. Traffic Lights

Submission details

Task:	Traffic Lights
Sender:	shreya01
Submission time:	2026-04-10 17:11:16 +0300
Language:	C++ (C++20)
Status:	READY
Result:	ACCEPTED

Test results

test	verdict	time	
#1	ACCEPTED	0.00 s	»
#2	ACCEPTED	0.00 s	»
#3	ACCEPTED	0.00 s	»
#4	ACCEPTED	0.00 s	»
#5	ACCEPTED	0.00 s	»
#6	ACCEPTED	0.30 s	»
#7	ACCEPTED	0.30 s	»
#8	ACCEPTED	0.62 s	»
#9	ACCEPTED	0.62 s	»
#10	ACCEPTED	0.62 s	»
#11	ACCEPTED	0.62 s	»
#12	ACCEPTED	0.63 s	»

Code

```cpp
#include <bits/stdc++.h>
using namespace std;

int main() {
    ios::sync_with_stdio(false);
    cin.tie(NULL);

    int x, n;
    cin >> x >> n;

    set<int> pos;
    multiset<int> len;

    pos.insert(0);
    pos.insert(x);
    len.insert(x);

    for(int i = 0; i < n; i++) {
        int p;
        cin >> p;

        auto it = pos.upper_bound(p);
        int r = *it;
        int l = *(--it);
        len.erase(len.find(r - l));

        len.insert(p - l);
        len.insert(r - p);
        pos.insert(p);
        cout << *len.rbegin() << " ";
    }
}
```